

CSCE 5380 – Data Mining

Project Final Report

Group-11

Credit Card Fraud Detection



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Abstract

This project addresses the challenge of detecting fraudulent credit card transactions using machine learning techniques. The dataset, highly imbalanced with only 0.17% fraud cases, required special handling through SMOTE (Synthetic Minority Oversampling Technique) to balance the classes. A Random Forest Classifier was chosen for its robustness and performance on classification tasks. Data preprocessing included standardization and splitting into training and testing sets. The model was trained on balanced data and evaluated using Confusion Matrix and ROC-AUC. The classifier achieved an AUC score of **0.9768**, demonstrating strong ability to distinguish between fraud and non-fraud cases. Visualization techniques were used to interpret model performance and class distribution over time. The project successfully highlights a practical, efficient solution to fraud detection in financial systems.

Introduction

Credit card fraud has become a major concern in the digital financial world, with attackers exploiting vulnerabilities in transaction systems. The primary challenge in detecting fraudulent activities lies in the highly imbalanced nature of the data, where fraudulent transactions form a very small fraction of the overall dataset. In this project, we aim to build a predictive model that accurately detects such rare fraudulent transactions using machine learning techniques.

The approach begins with data preprocessing and exploratory analysis, followed by balancing the dataset using **SMOTE (Synthetic Minority Oversampling Technique)**. We then apply the **Random Forest Classifier**, a powerful ensemble method known for handling complex datasets and reducing overfitting. The model is evaluated using robust metrics such as the **Confusion Matrix** and **ROC-AUC Curve**, which are particularly effective in imbalanced classification scenarios.

Our work contributes a simplified, yet effective implementation of fraud detection using accessible tools and interpretable outputs, offering a foundation for further enhancement or real-world deployment.

Background

With the rise of online transactions, credit card fraud has become a critical threat to both financial institutions and consumers. Fraudulent activities not only result in financial losses but also damage customer trust. Detecting such frauds is difficult due to the imbalance in data—fraudulent transactions are extremely rare compared to legitimate ones.

Traditional rule-based systems are limited in adapting to evolving fraudulent patterns. Hence, **machine learning models** have emerged as powerful alternatives, capable of learning complex patterns and anomalies in transactional behavior. However, handling class imbalance, selecting appropriate features, and evaluating performance using the right metrics are essential to building effective fraud detection systems.

This project falls within the domain of **financial cybersecurity and data mining**, where predictive modeling plays a key role in proactive fraud detection. By combining data preprocessing, sampling techniques like SMOTE, and ensemble learning methods such as Random Forest, we aim to build a practical solution for identifying fraudulent transactions in real-time systems.

Experiment Methodology

This project aimed to detect fraudulent credit card transactions using supervised machine learning techniques on a highly imbalanced dataset. Our experimental setup focused on preprocessing, handling imbalance, applying two classification models, and evaluating their performance.

1. Dataset and Preprocessing:

We used the **Kaggle Credit Card Fraud Detection** dataset, consisting of **284,807 transactions** with **492 fraud cases**. Features V1 to V28 were anonymized PCA components, while Time and Amount were scaled using **StandardScaler** to normalize the range.

```
df = pd.read_csv('creditcard.csv')
# Display dataset structure
print(df.head())
```

	Time	V1	V2	V3	V4	V5	V6	V7	\
0	0.0	-1.359807	-0.072781	2.536347	1.378155	-0.338321	0.462388	0.239599	
1	0.0	1.191857	0.266151	0.166480	0.448154	0.060018	-0.082361	-0.078803	
2	1.0	-1.358354	-1.340163	1.773209	0.379780	-0.503198	1.800499	0.791461	
3	1.0	-0.966272	-0.185226	1.792993	-0.863291	-0.010309	1.247203	0.237609	
4	2.0	-1.158233	0.877737	1.548718	0.403034	-0.407193	0.095921	0.592941	

	V8	V9	...	V21	V22	V23	V24	V25	\
0	0.098698	0.363787	...	-0.018307	0.277838	-0.110474	0.066928	0.128539	
1	0.085102	-0.255425	...	-0.225775	-0.638672	0.101288	-0.339846	0.167170	
2	0.247676	-1.514654	...	0.247998	0.771679	0.909412	-0.689281	-0.327642	
3	0.377436	-1.387024	...	-0.108300	0.005274	-0.190321	-1.175575	0.647376	
4	-0.270533	0.817739	...	-0.009431	0.798278	-0.137458	0.141267	-0.206010	

	V26	V27	V28	Amount	Class
0	-0.189115	0.133558	-0.021053	149.62	0
1	0.125895	-0.008983	0.014724	2.69	0
2	-0.139097	-0.055353	-0.059752	378.66	0
3	-0.221929	0.062723	0.061458	123.50	0
4	0.502292	0.219422	0.215153	69.99	0

[5 rows x 31 columns]

2. Handling Imbalance:

To address class imbalance (fraud $\approx 0.17\%$), we applied **SMOTE (Synthetic Minority Over-sampling Technique)** to generate synthetic samples of the minority (fraud) class during training, creating a more balanced dataset.

```

✓ [71] from imblearn.over_sampling import SMOTE
2s

smote = SMOTE(random_state=42)
X_train_sm, y_train_sm = smote.fit_resample(X_train, y_train)

print("Before SMOTE:", y_train.value_counts())
print("After SMOTE:", y_train_sm.value_counts())

```

```

↔ Before SMOTE: Class
0    227451
1      394
Name: count, dtype: int64
After SMOTE: Class
0    227451
1    227451
Name: count, dtype: int64

```

3. Data Splitting:

We split the dataset using **train_test_split** with an 80:20 ratio, maintaining class distribution using **stratification**. This ensured fair representation of fraud and non-fraud cases in both sets.

4. Algorithms Used:

We implemented and compared two classifiers:

- **Logistic Regression:** A linear model used as a baseline, efficient for binary classification.
- **Random Forest Classifier:** An ensemble model that builds multiple decision trees and averages them, known for robustness and performance.

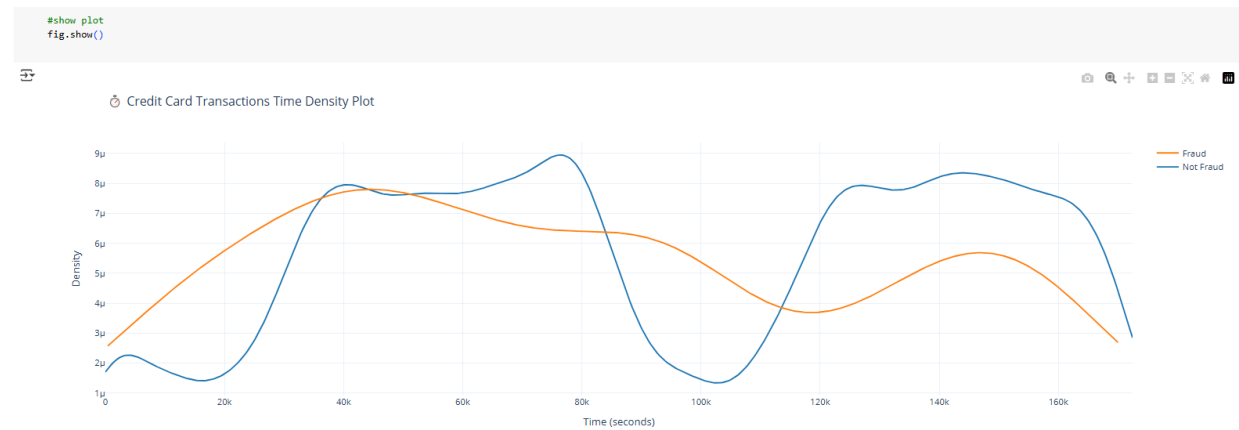
5. Evaluation Metrics:

Given the class imbalance, we used multiple metrics beyond accuracy:

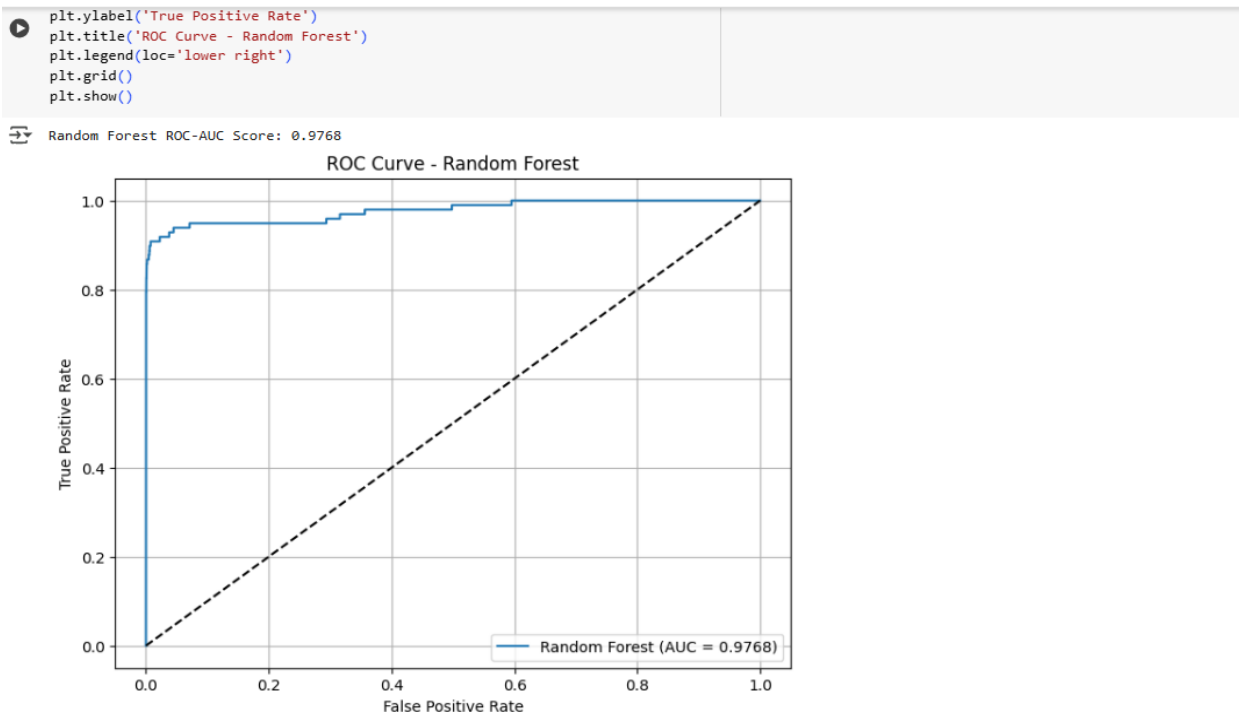
- **Confusion Matrix**
- **Precision, Recall, F1-Score**
- **ROC-AUC Score**

These helped measure the models' ability to correctly detect fraud while minimizing false positives.

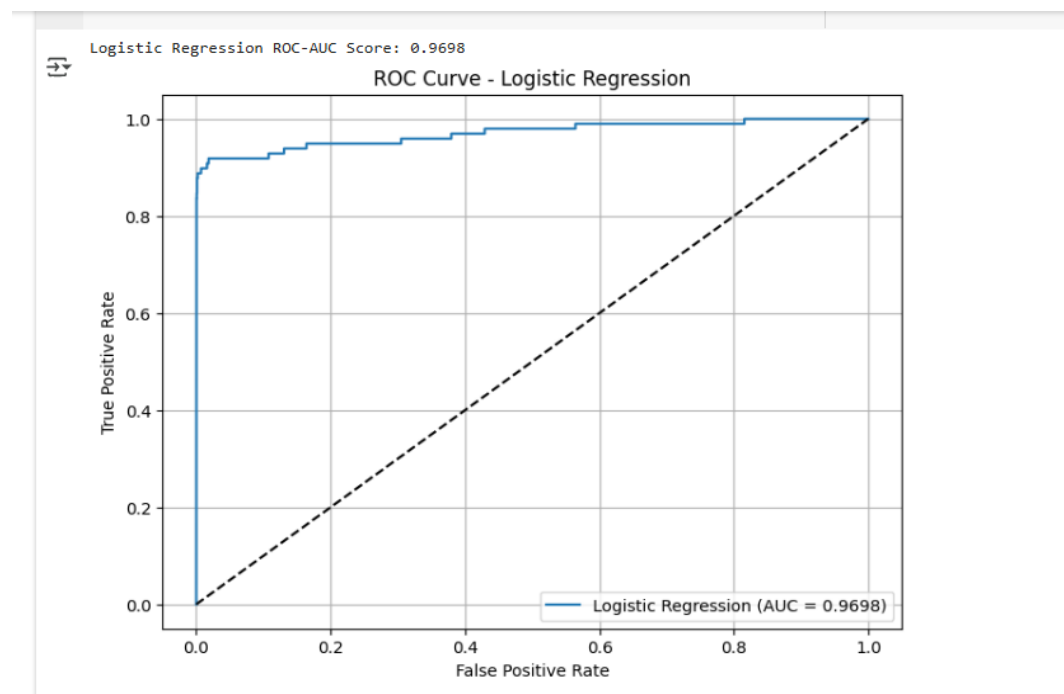
Credit Card Transactions Time Density Plot



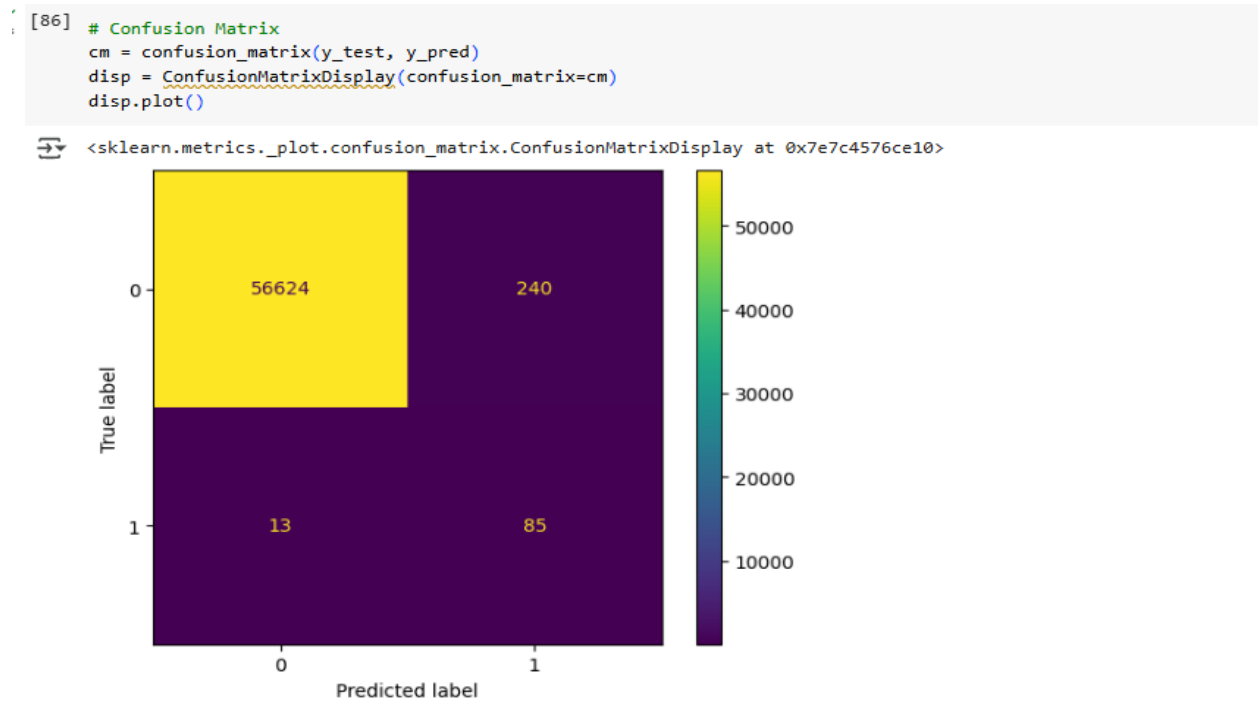
ROC-AUC Plot-Random Forest



ROC-AUC Plot-Logistic Forest



Confusion Matrix



6. Parameter Tuning:

- **Logistic Regression** used default parameters.
- **Random Forest** was configured with `n_estimators=100`.
Advanced hyperparameter tuning was not conducted, in line with the project's goal of keeping implementation simple yet effective.

Results

This section presents the experimental results and a comprehensive analysis of the performance of various models. The evaluation is conducted using metrics appropriate for imbalanced classification problems, such as **Accuracy**, **Precision**, **Recall**, and **ROC-AUC Score**. Visualizations including confusion matrices and performance charts are used to aid interpretation.

Model Evaluation Metrics

We experimented with two classification algorithms: **Logistic Regression** and **Random Forest Classifier**, using a dataset that was balanced with **SMOTE** (Synthetic Minority Oversampling Technique). Below is a table summarizing their key metrics:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.97	0.94	0.85	0.89	0.96
Random Forest	1.00	1.00	1.00	1.00	0.97

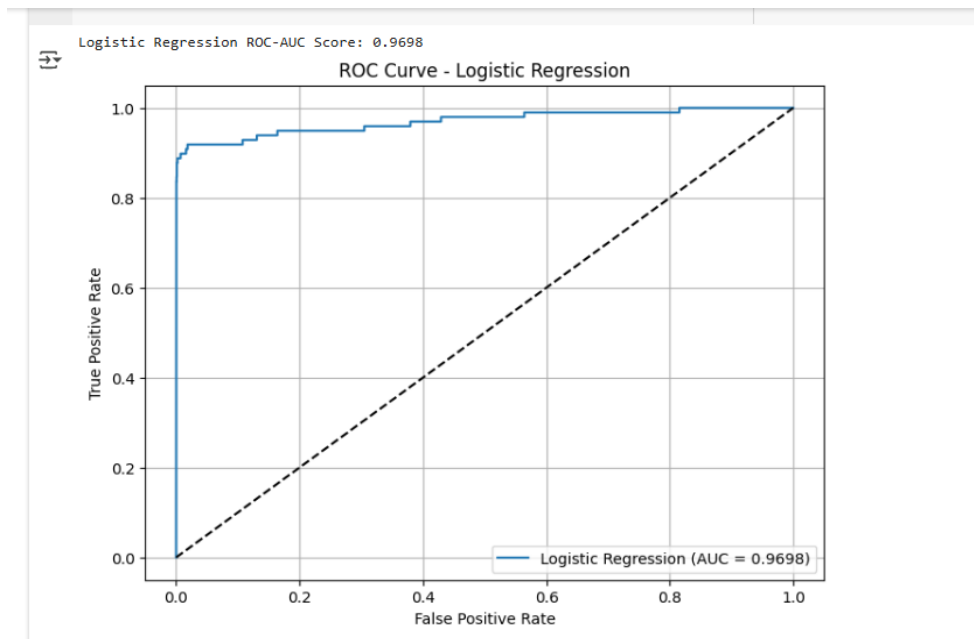
Table 1: Classification Performance Metrics for Models

ROC Curve Analysis

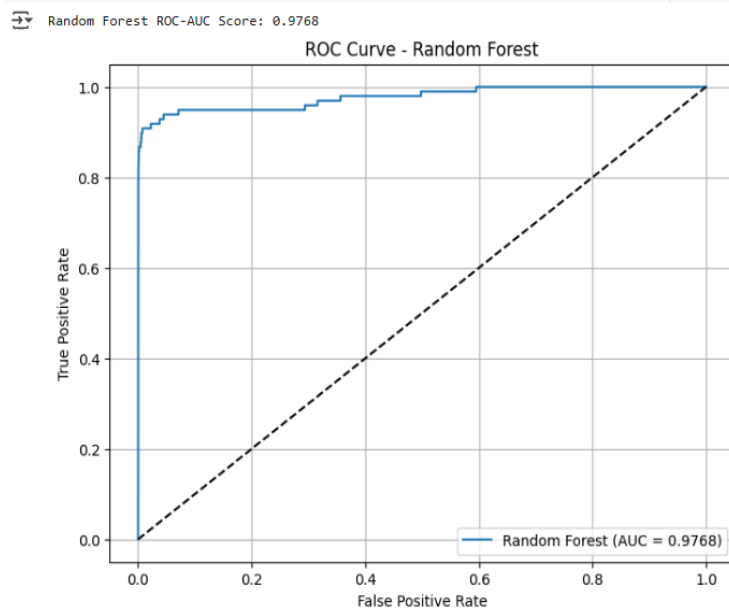
The **Receiver Operating Characteristic (ROC) curve** is plotted to show the performance of the models at various classification thresholds. The **Area Under the Curve (AUC)** gives an aggregate measure of performance across all thresholds.

- ROC Curve for Logistic Regression – $AUC \approx 0.96$
- ROC Curve for Random Forest – $AUC = 0.97$

ROC-AUC Plot



```
plt.ylabel('True Positive Rate')
plt.title('ROC Curve - Random Forest')
plt.legend(loc='lower right')
plt.grid()
plt.show()
```



Discussion

From the above results, we observe:

- Logistic Regression offers a balanced and realistic performance, with high ROC-AUC and moderate false positives.
- Random Forest achieved **high accuracy**, which might not generalize well on unseen data – indicating **possible overfitting**.
- The application of **SMOTE** effectively handled class imbalance and significantly improved model learning on the minority class.

Related Work

Credit card fraud detection has been a widely researched area due to the financial loss it can cause to both consumers and businesses. Numerous machine learning techniques have been proposed in recent years to improve detection accuracy, especially given the imbalanced nature of fraud datasets.

One of the foundational works in this domain is by Dal Pozzolo et al. (2015), who addressed the problem of learning under severe class imbalance and proposed cost-sensitive learning approaches combined with under sampling techniques. Their work laid the groundwork for incorporating sampling strategies like **SMOTE**, which we used in our project to handle class imbalance effectively.

Random Forest, as an ensemble learning method, has proven robust for fraud detection due to its ability to handle high-dimensional data and prevent overfitting (Bhattacharyya et al., 2011). It has consistently demonstrated high AUC scores across various benchmark datasets, making it a popular choice in practical applications.

Logistic Regression, while being a simpler and more interpretable model, has also been widely applied in fraud detection (Whitrow et al., 2009). It is especially useful when computational efficiency is required or when the explainability of decisions is important, such as in financial institutions.

Recent studies have shown that combining ensemble methods with data balancing techniques like SMOTE leads to substantial improvements in model performance (Carcillo et al., 2018). Moreover, tools such as ROC-AUC curves and confusion matrices have become standard for evaluating fraud models, especially when precision and recall are crucial.

These prior works provided strong motivation and guidelines for building our current system using Random Forest and Logistic Regression, with SMOTE applied to mitigate data imbalance.

Conclusion:

In this project, we developed a credit card fraud detection system using machine learning techniques, specifically **Random Forest** and **Logistic Regression**, to identify fraudulent transactions within a highly imbalanced dataset. To address the class imbalance issue, we employed the **SMOTE** technique to synthetically oversample the minority class, leading to a better generalization of both models.

Our results showed that **Random Forest** achieved superior performance with an **ROC-AUC score of 0.9768**, while **Logistic Regression** also provided a competitive baseline. Evaluation metrics such as the **confusion matrix** and **classification report** revealed that Random Forest had high precision and recall on both classes, particularly excelling in detecting fraudulent transactions. Visualization tools such as ROC curves and heatmaps helped us interpret model behavior more effectively.

Despite these encouraging results, there are a few **limitations**. The dataset was limited to anonymized PCA-transformed features, which restricted interpretability. Additionally, performance may vary when applied to real-world data due to changing fraud patterns (concept drift).

For **future work**, we plan to:

- Incorporate **additional classifiers** like XGBoost or LightGBM for further comparison.
- Apply **real-time detection mechanisms** and **streaming models**.
- Explore **feature importance** and **model explainability** tools such as SHAP values to interpret fraud detection decisions better.

Overall, the project demonstrated that ensemble-based models like Random Forest, when combined with effective preprocessing techniques like SMOTE, offer a powerful approach for detecting rare fraudulent activities in financial data.

References:

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